



CEO Turnover and Executive Pay Dispersion in Europe

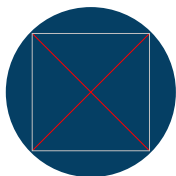
Achmad Rizky Akbar and Kajetan Zduńczyk

TU Dortmund



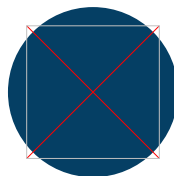
- 1 **Motivation and Economic Relevance**
Preparation of Final Analysis Panel and Pay Dispersion
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- 3 **Data Visualizations**
- 4 **Methodology**
- 5 **Results**
- 6 **Limitations and Synthesis**

Motivation, Research Question, and Economic Relevance



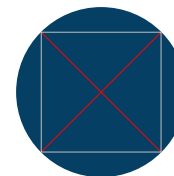
Motivation

A CEO turnover is a major corporate governance event that can reshape a firm's strategic direction, leadership priorities, and internal incentive structure. Because CEO succession often coincides with renewed compensation decisions, turnover may affect both the level of CEO pay and the relative pay gap between the CEO and other senior executives.



Research Question

This study examines whether CEO turnover causally changes CEO compensation and CEO-to-executive pay dispersion within European firms.



Economic Relevance

CEO turnover may reset compensation incentives and reshape pay inequality within top management. By examining both CEO pay and CEO-to-executive pay dispersion, the study shows whether leadership transitions affect not only how much CEOs are paid, but also how rewards are distributed among senior executives.



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Motivation and Economic Relevance

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Data Visualizations

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Methodology

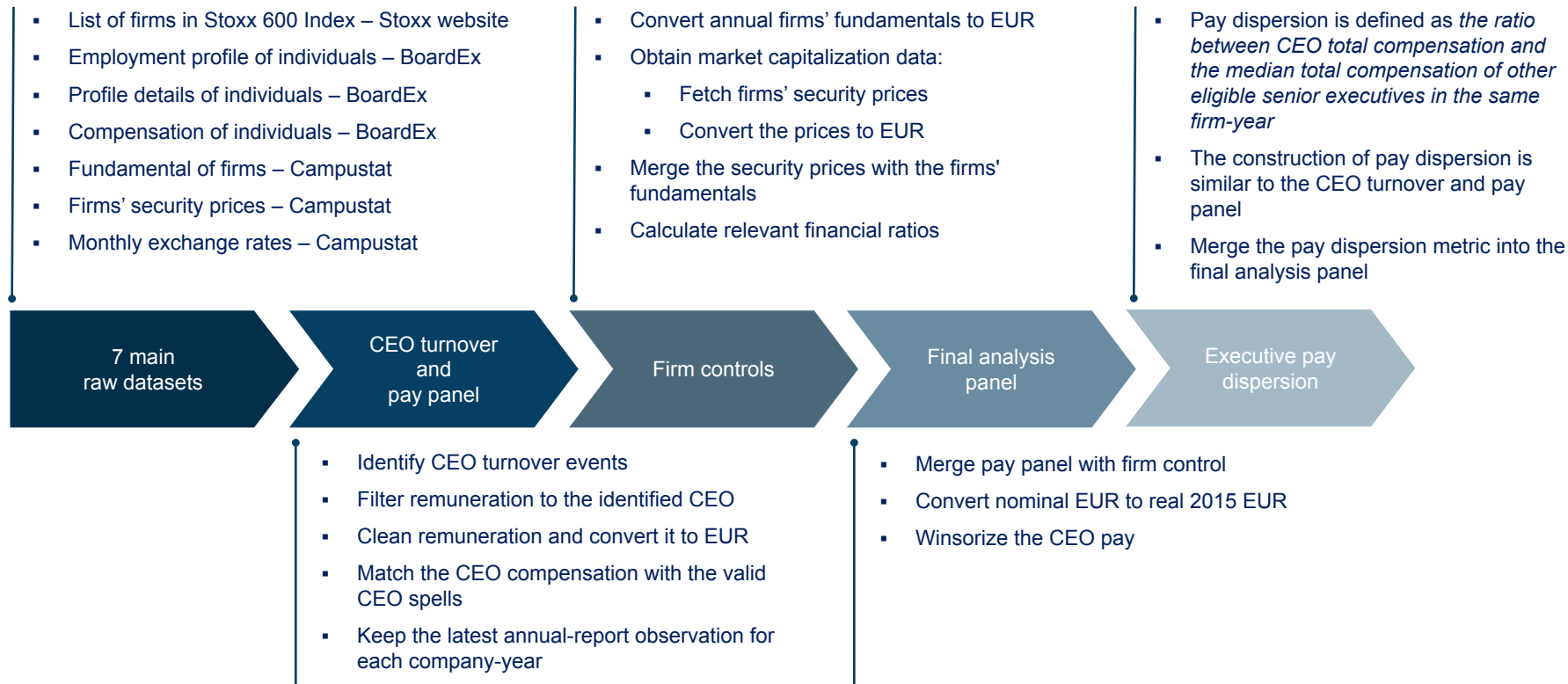
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Results

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Limitations and Synthesis

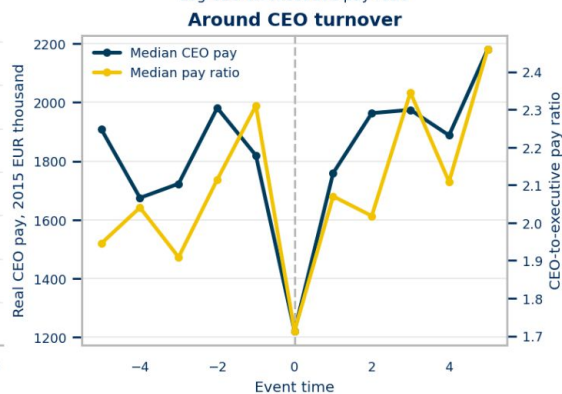
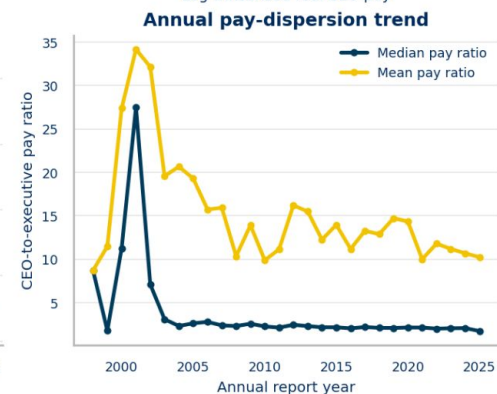
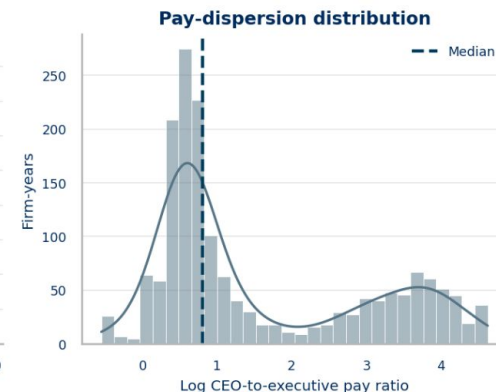
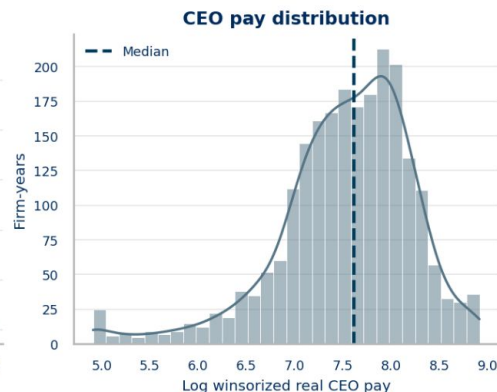
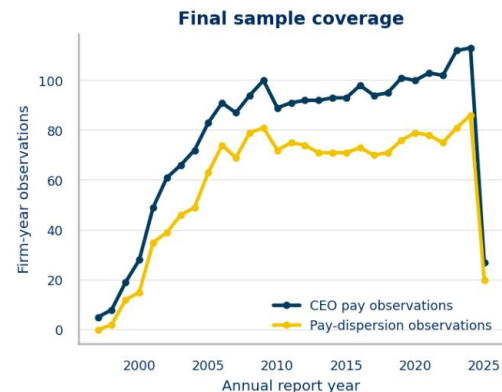
Preparation of Final Analysis Panel and Pay Dispersion





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Data Visualizations





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Causal Inference

- Goal: estimate effect of CEO turnover
- Outcomes:
 - log CEO pay
 - log CEO-to-executive pay dispersion
- Treatment:
 - CEO turnover and post-turnover period
- Method:
 - Difference-in-Differences
 - event-study around turnover year
- Controls:
 - firm size, ROA, leverage, country, sector, year fixed effects

Supervised Learning

- Goal: predict pay, turnover, and dispersion
- Tasks:
 - CEO pay regression
 - turnover classification
 - pay-dispersion regression
- Design:
 - train: pre-2020
 - test: 2020-2025
- Models:
 - Ridge / Logistic baseline
 - Random Forest, XGBoost

Unsupervised Learning

- Goal: identify compensation profiles
- Inputs:
 - pay, dispersion, firm fundamentals
 - turnover indicators
 - country, sector, year
- Method:
 - preprocessing pipeline
 - PCA
 - K-Means clustering
- Validation:
 - silhouette score
 - cluster profile comparison



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Causal Inference

- Main finding:
 - CEO turnover compresses pay dispersion but does not permanently change CEO pay.
- CEO pay: no significant average post-turnover effect
- DiD = -0.060, $p = 0.173$
- Turnover-year pay drops temporarily
- Event $\tau = 0$: -0.532
- Pay dispersion falls significantly after turnover
- DiD = -0.339, $p < 0.001$

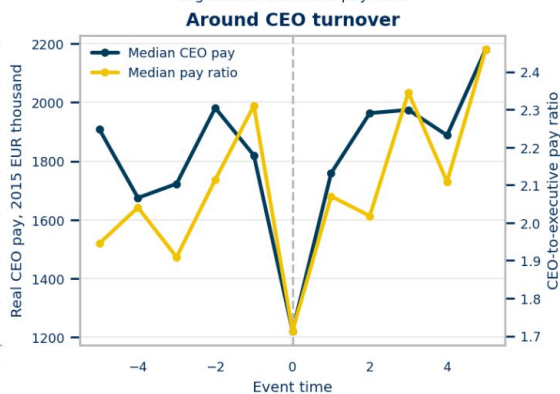
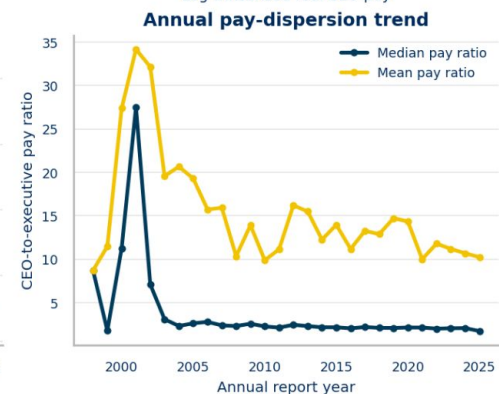
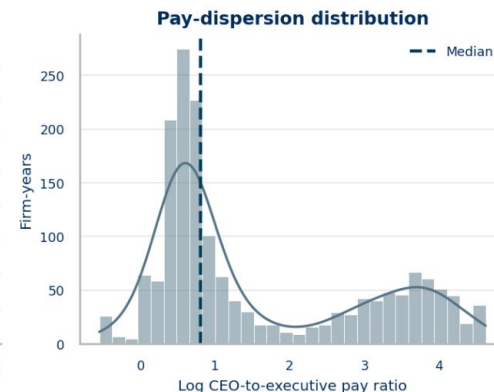
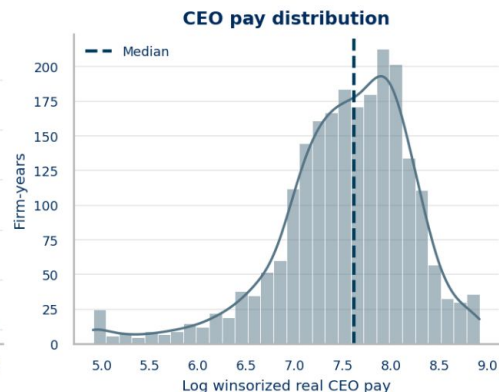
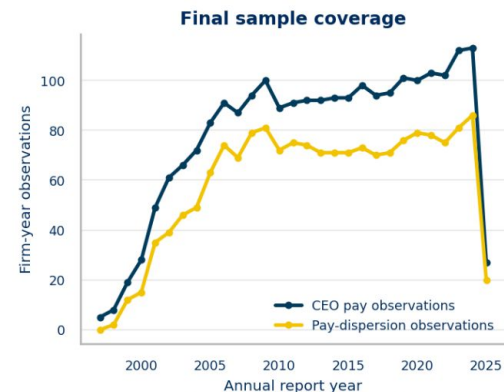
Supervised Learning

- Main finding:
 - Pay and pay dispersion are moderately predictable
 - turnover is difficult to predict.
- CEO pay prediction: best model XGBoost
 - RMSE =
 - $R^2 = 0.413$
- Turnover classification: weak predictability
 - ROC-AUC = 0.629
 - MCC = 0.124
- Pay dispersion prediction: best model Random Forest
 - RMSE =
 - $R^2 = 0.404$

Unsupervised Learning

- Main finding:
 - Clusters separate firm-years by compensation structure and turnover timing.
- Best solution: $k = 3$
- Clusters broadly distinguish:
 - pre-turnover, turnover-year, and post-turnover/high-pay profiles
- Useful descriptively, not causal evidence

Results





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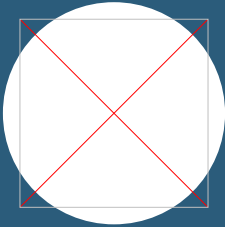
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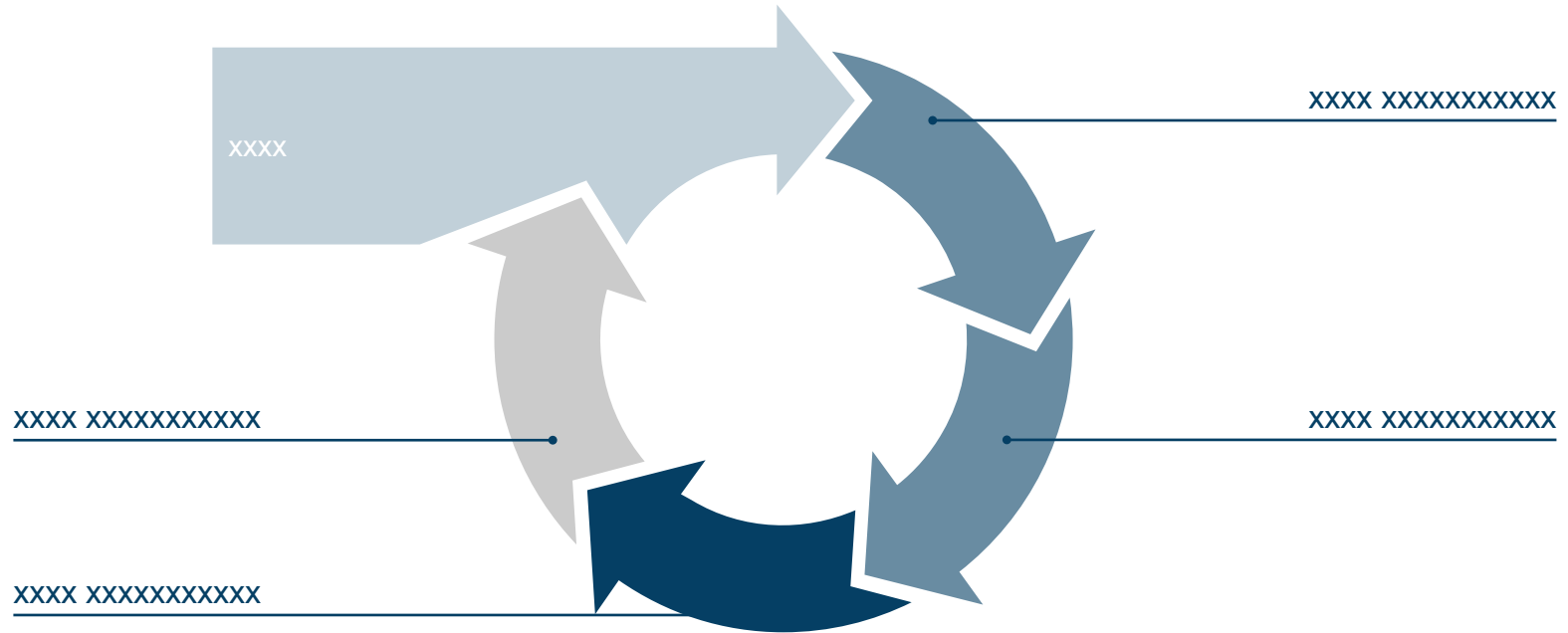


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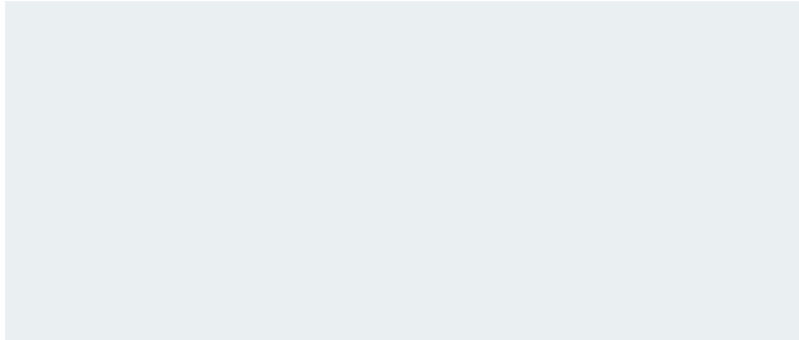
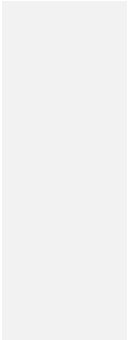
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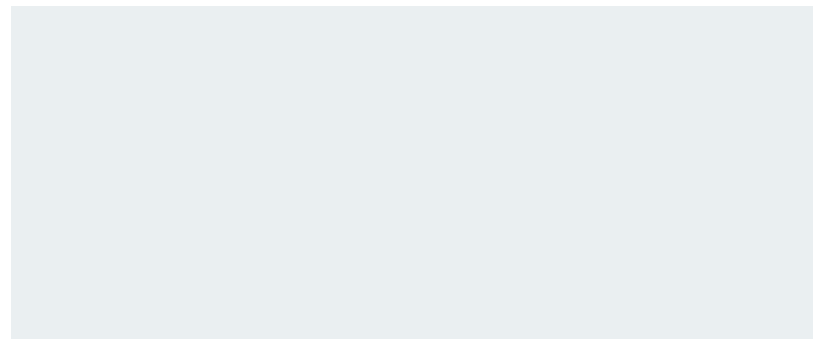
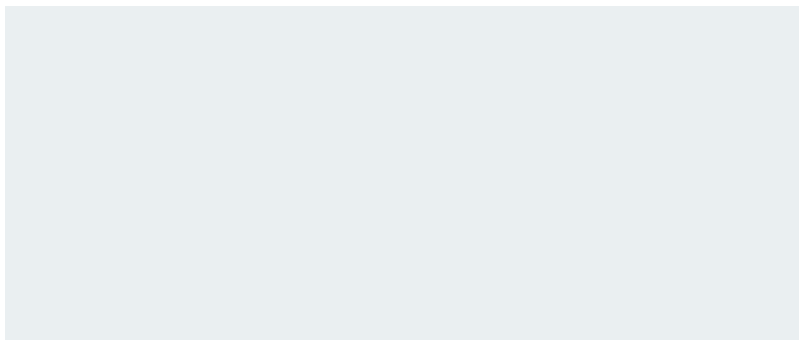
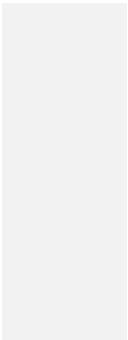
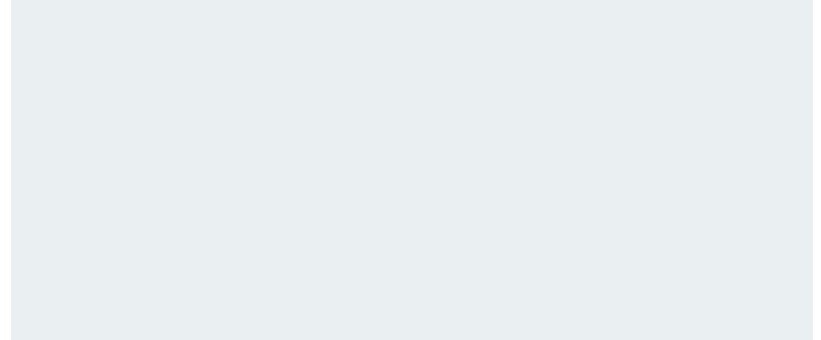




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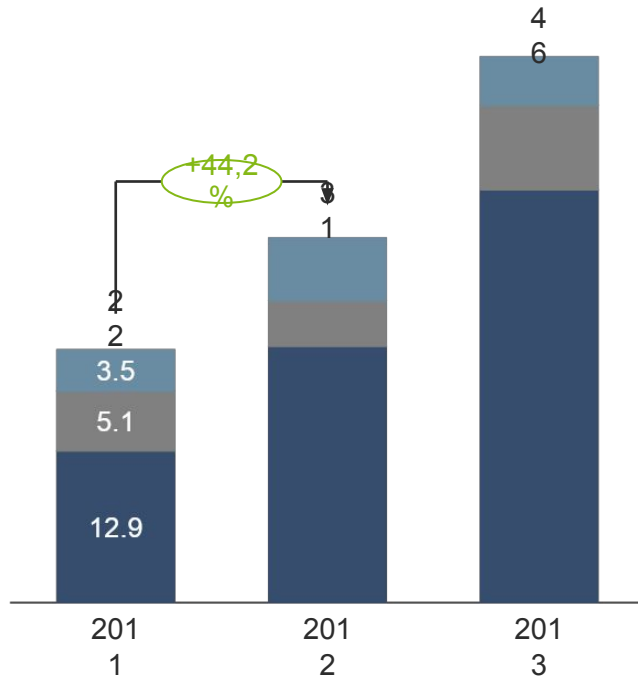




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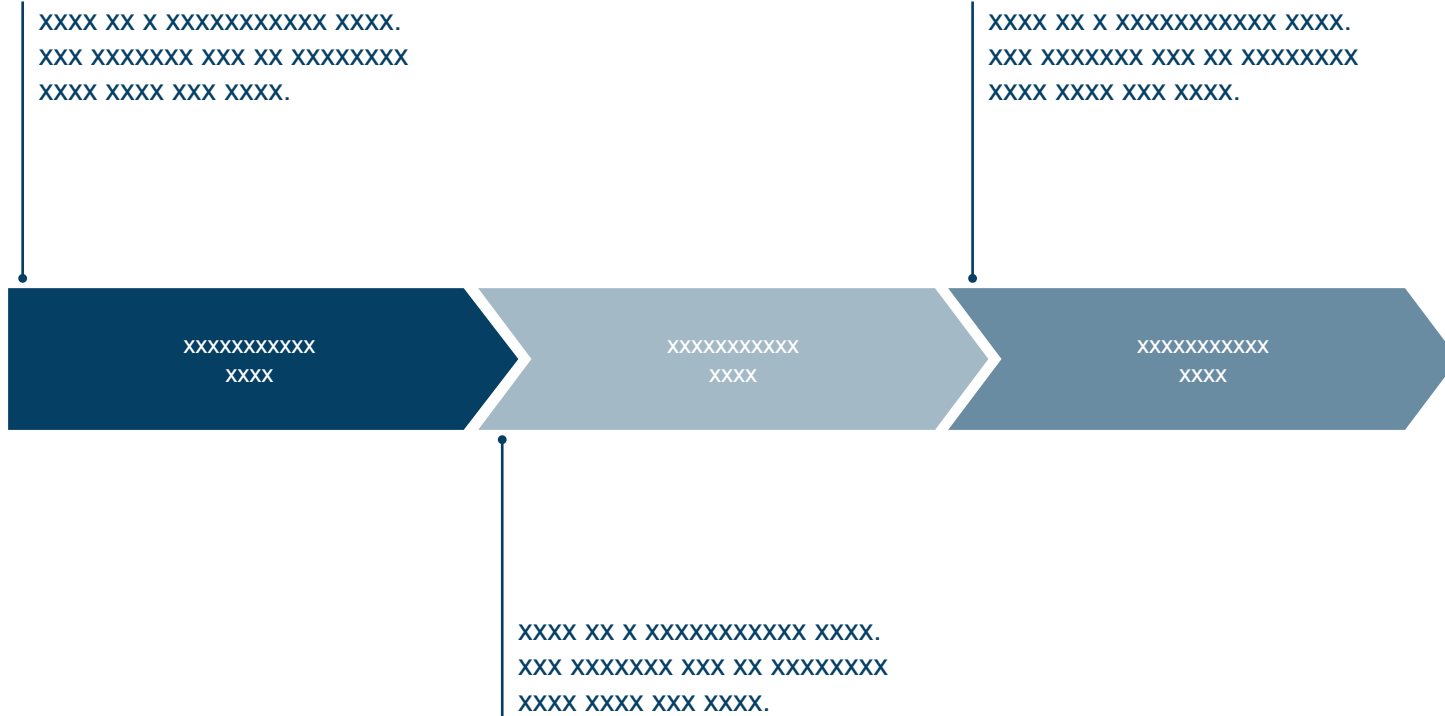
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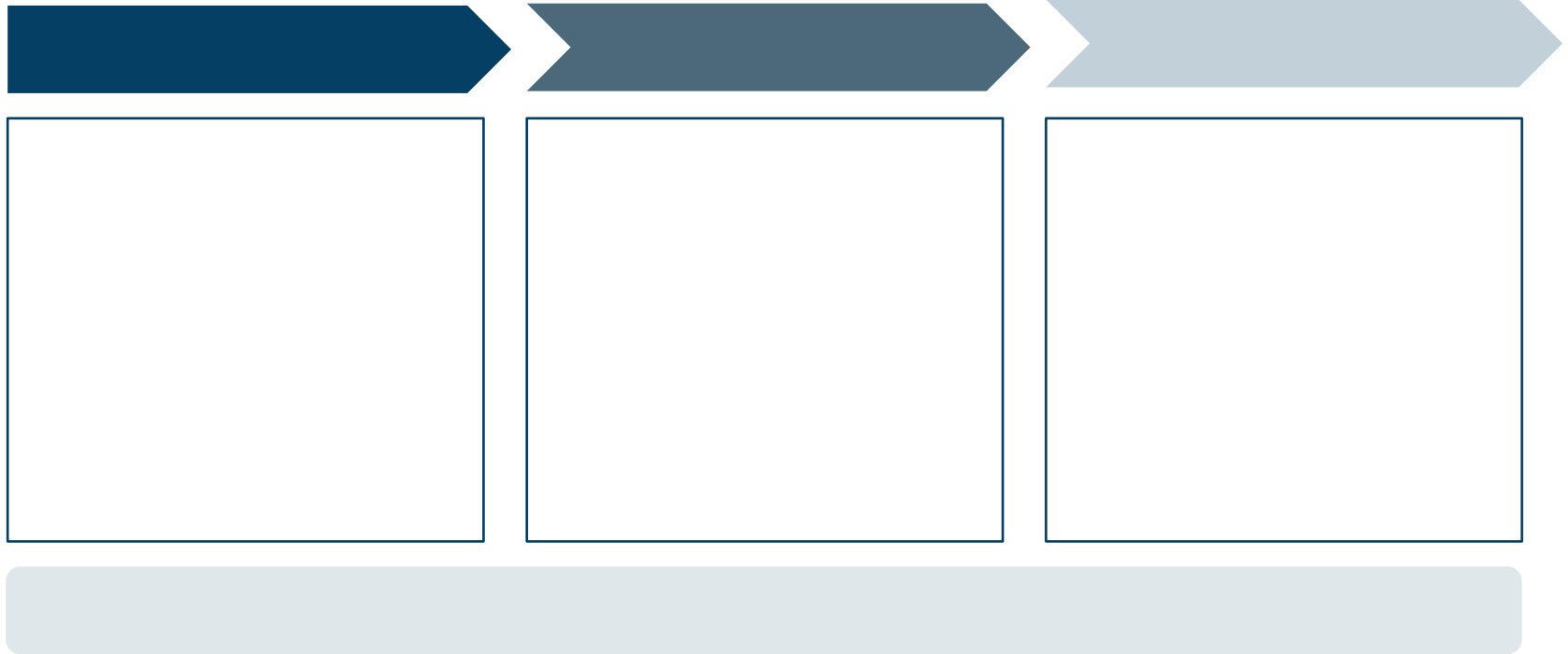
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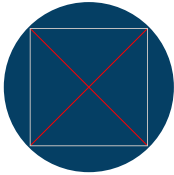


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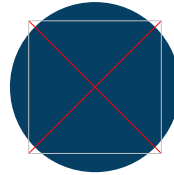






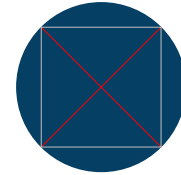
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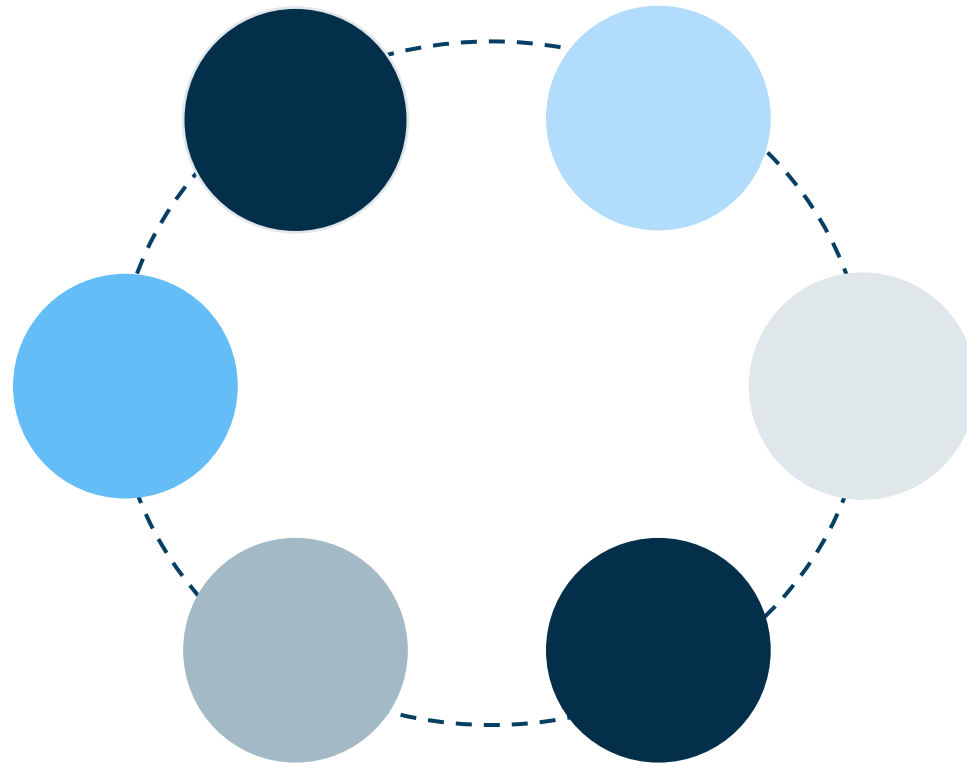
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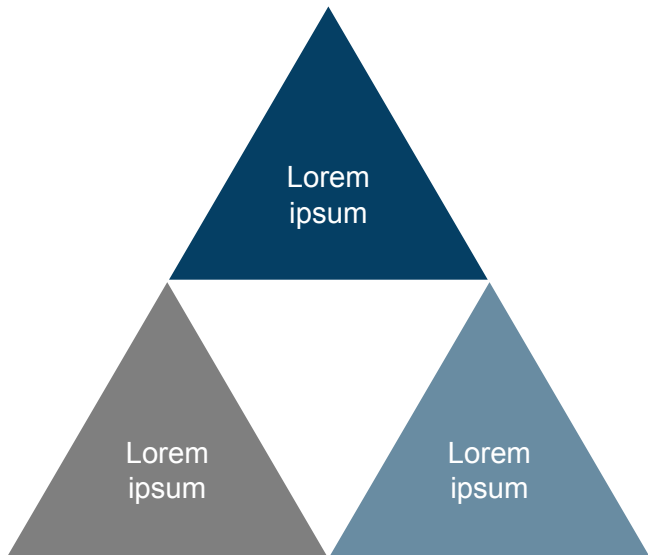
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